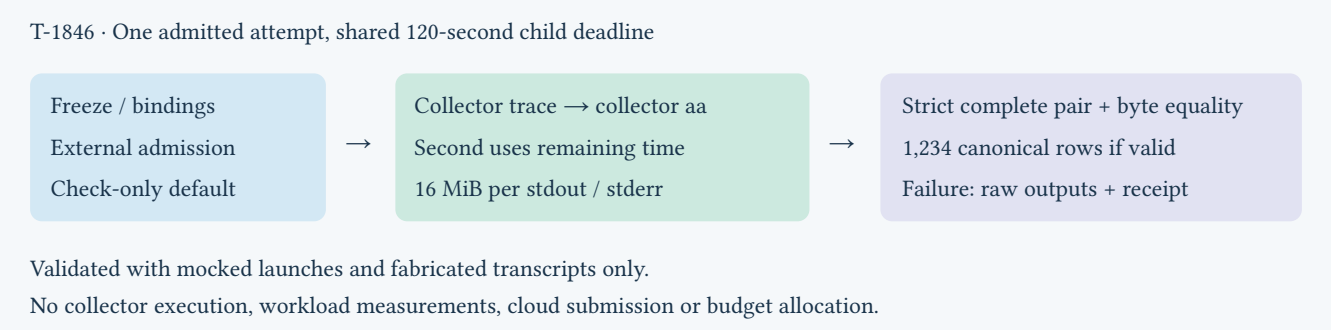
Paired source-quotient capture and export

T-1846 · 2026-11-07 · Software adapter validation

Implemented the execution/export path for the source-Fp quotient trace. Two independent collector processes belong to one admitted attempt, with a shared child deadline and exact A/A transcript equality. Successful export retains only quotient-stage counts; scalar-field arithmetic, other source phases and valid wall-clock claims remain outside this measurement boundary.



Capture. Check frozen protocol, all source/adapter/binary bindings and external admission before execution. Default checks only. One exclusive attempt directory, no retries. The second process gets the remaining portion of 120 seconds. Child stdout/stderr each have a 16 MiB file limit, respecting tighter inherited limits. External job enforcement remains responsible for CPU/RAM and total attempt lifetime. Evidence JSON does not authenticate itself or grant cloud authority.

Failure custody. Timeout, nonzero exit or launch error stops the pair. Raw output, stderr and structured receipt remain; unstarted processes are not labelled executed. Complete but malformed or unequal transcripts become invalid data. Missing host-termination receipts never imply success. Failed attempts produce no fabricated target rows or successful exports.

Final software validation	Result
Mocked/synthetic test methods	7 pass
Complete fabricated pair	1,234 canonical rows
Upstream export check	Pass with file verification
Shared deadline exhausted	No second launch
Prelaunch deadline exhausted	No child launch
Failure and unequal-pair controls	Receipts retained; export refused
Protocol draft	Valid; not frozen/admitted
Actual workload collection	Not executed

Canonical export. Two arms, 617 instances each, seed zero, one replicate and no warmups. Per-target quotient/product/inversion counts, exact raw-file hashes and shared receipt. Clock validity is false; verification is no_claim because producer checks are not independent review. Existing and renamed attempt paths, changed bindings and tampered outputs are rejected.

Evidence. validation-01 passed six methods; final validation-02 adds file-limit and prelaunch tests and distinguishes attempted from executed on launch failure. No failed test runs. Prior capture variant retained. PCC-20261107-894d19 is unsigned producer evidence. README.md defines commands, provenance and limitations. Original 4,563-case cohort unchanged; no cloud or full Cargo rerun.

Next. Bind an admitted production build/preflight, reconcile actual remaining funds/capacity, freeze and publish the protocol before workload data. This turn completes the adapter and its software checks; source quotient frequencies and aggregate savings remain unmeasured. Sources: predecessor trace and resultant adapters, pinned canonical protocol docs/schema; KB unavailable.